Exp. No: 5

Electric Potential Difference between Two Points in Free Space

Aim:

To write program for Electric potential difference between two points in free space medium using MATLAB.

Objectives:

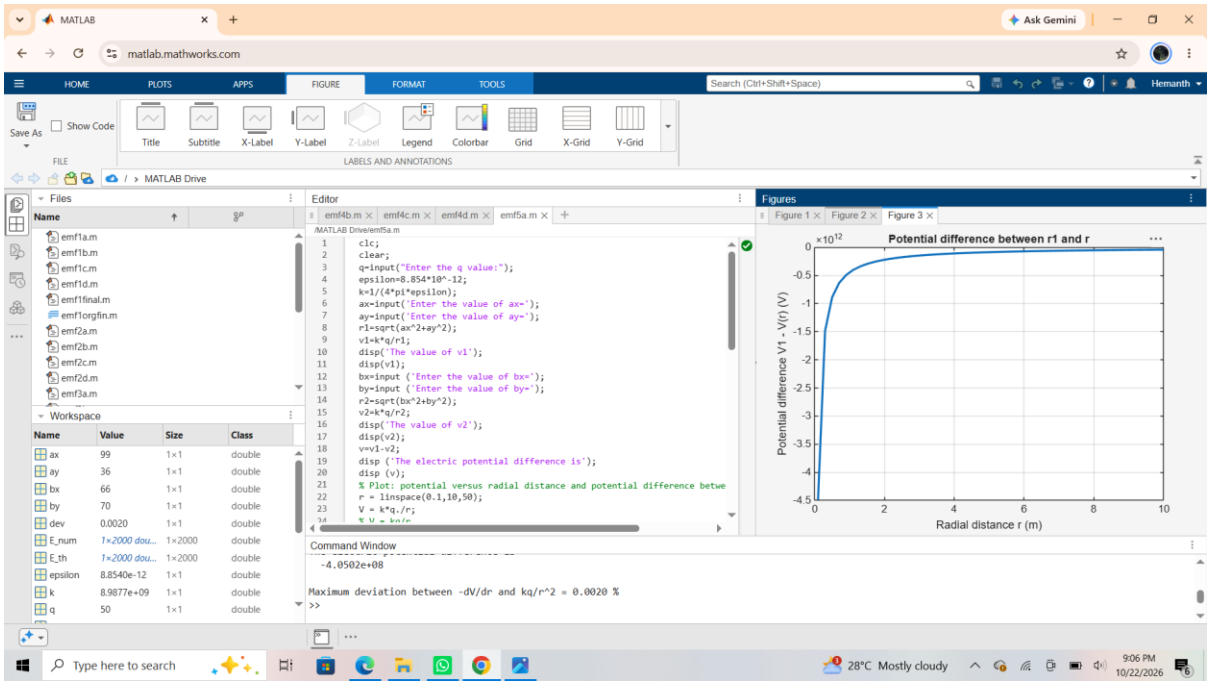
1. To determine the electric potential difference between two points separated by a distance R in free space.
2. To plot the variation of potential difference with separation distance.
3. To relate the potential gradient to electric field intensity for validation.

Test case (i) Determination of electric potential difference between two points for a given charge

Program:

```
clc;
clear;
q=input("Enter the q value:");
epsilon=8.854*10^-12;
k=1/(4*pi*epsilon);
ax=input('Enter the value of ax=');
ay=input('Enter the value of ay=');
r1=sqrt(ax^2+ay^2);
v1=k*q/r1;
disp('The value of v1');
disp(v1);
bx=input ('Enter the value of bx=');
by=input ('Enter the value of by=');
```

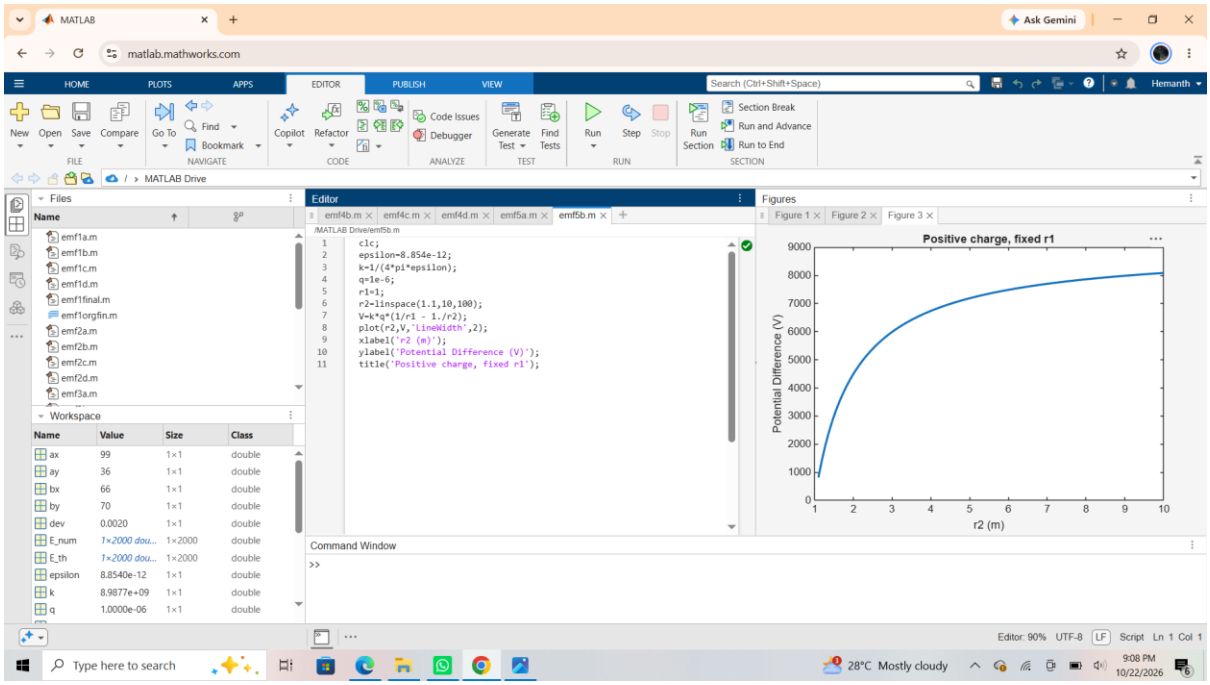
```
r2=sqrt(bx^2+by^2);
v2=k*q/r2;
disp('The value of v2');
disp(v2);
v=v1-v2;
disp ('The electric potential difference is');
disp (v);
% Plot: potential versus radial distance and potential difference between r1 and r
R=  linspace(0.1,10,50);
V = k*q./r;
% V = kq/r figure; plot(r,V,'LineWidth',2); hold on;
plot(r1,v1,'ro','MarkerFaceColor','r'); grid on;
xlabel('Radial distance r (m)'); ylabel('Potential V (V)');
title('Potential V = kq/r'); legend('V(r)','V1 at r1');
figure;
plot(r,k*q*(1/r1 - 1./r),'LineWidth',2); grid on;
xlabel('Radial distance r (m)'); ylabel('Potential difference V1 - V(r) (V)');
title('Potential difference between r1 and r');
% Independent check: E = -dV/dr (numerical gradient) compared with E = kq/r^2
rc = linspace(1,10,2000);
E_num = -gradient(k*q./rc,rc);
E_th = k*q./rc.^2;
dev = max(abs(E_num(2:end-1) - E_th(2:end-1))./E_th(2:end-1))*100;
fprintf('Maximum deviation between -dV/dr and kq/r^2 = %.4f %%\n', dev);
```



Test case (ii) A positive charge with fixed r_1 = 1 m and r_2 varying.

Program:

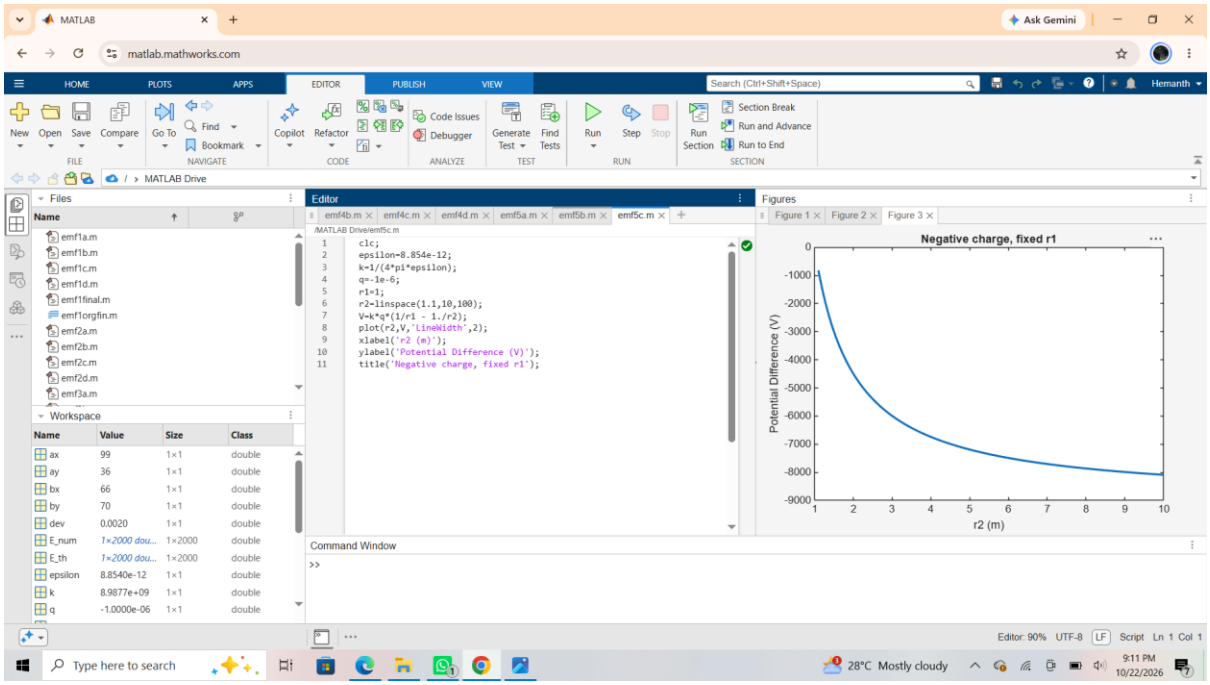
```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=1;
r2=linspace(1.1,10,100);
V=k*q*(1/r1 - 1./r2);
plot(r2,V,'LineWidth',2);
xlabel('r2 (m)');
ylabel('Potential Difference (V)');
title('Positive charge, fixed r1');
```



Test case (iii) A negative charge with fixed r_1 and varying r_2:

Program:

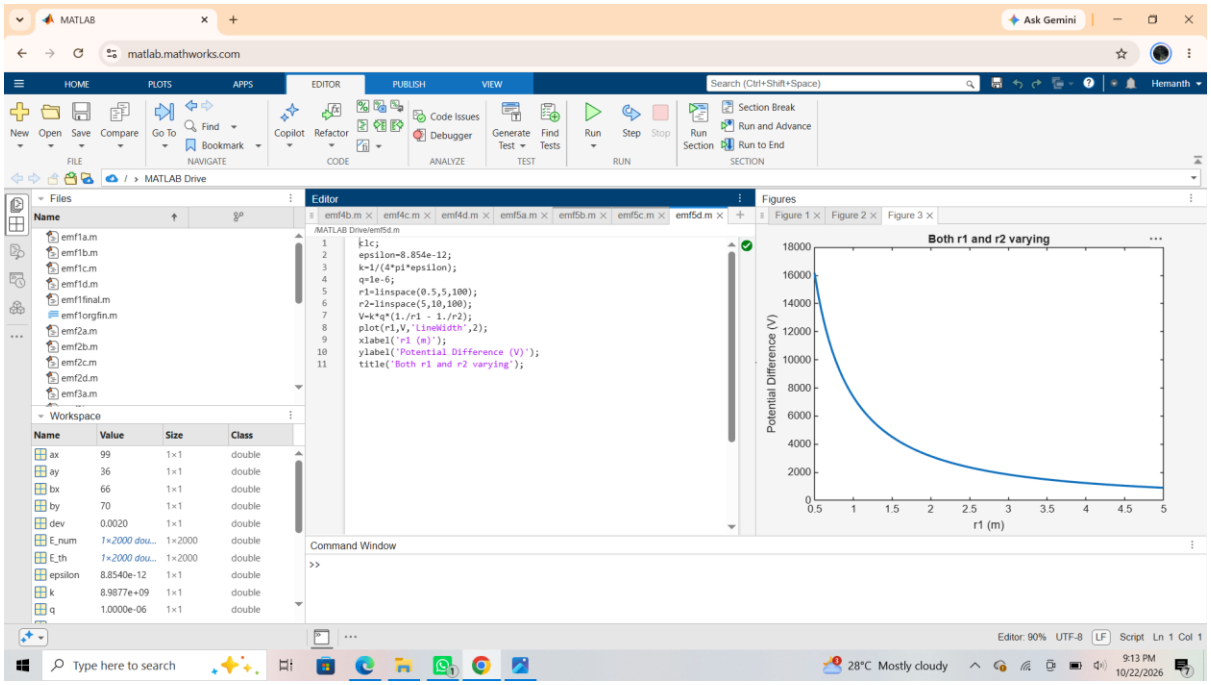
```
clc;
epsilon=8.854e-12;
 k=1/(4*pi*epsilon);
q=-1e-6;
 r1=1;
 r2=linspace(1.1,10,100);
 V=k*q*(1/r1 - 1./r2);
 plot(r2,V,'LineWidth',2);
 xlabel('r2 (m)');
 ylabel('Potential Difference (V)');
title('Negative charge, fixed r1');
```



Test case (iv) Both r_1 and r_2 varying with positive charge 3:

Program:

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=linspace(0.5,5,100);
r2=linspace(5,10,100);
V=k*q*(1./r1 - 1./r2);
plot(r1,V,'LineWidth',2);
xlabel('r1 (m)');
ylabel('Potential Difference (V)');
title('Both r1 and r2 varying');
```


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matlab.mathworks.com
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Code Issues
Debugger
Generate Test
Find Tests
Run
Step
Stop
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Section Break
Run and Advance
Run to End
FILE
NAVIGATE
CODE
ANALYZE
TEST
RUN
SECTION
MATLAB Drive
Files
Name
emf1a.m
emf1b.m
emf1c.m
emf1d.m
emf1final.m
emf1orgfin.m
emf2a.m
emf2b.m
emf2c.m
emf2d.m
emf3a.m
Workspace
Name
Value
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ax 99 1×1 double
ay 36 1×1 double
bx 66 1×1 double
by 70 1×1 double
dev 0.0020 1×1 double
E_num 1×2000 dou... 1×2000 double
E_th 1×2000 dou... 1×2000 double
epsilon 8.8540e-12 1×1 double
k 8.9877e+09 1×1 double
q 1.0000e-06 1×1 double
Editor
emf4b.m
emf4c.m
emf4d.m
emf5a.m
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emf5c.m
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clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=linspace(0.5,5,100);
r2=linspace(5,10,100);
V=k*q*(1./r1 - 1./r2);
plot(r1,V,'LineWidth',2);
xlabel('r1 (m)');
ylabel('Potential Difference (V)');
title('Both r1 and r2 varying');
Figures
Figure 1
Figure 2
Figure 3
Both r1 and r2 varying
Potential Difference (V)
r1 (m)
Command Window
>>
Editor: 90%
UTF-8
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Script
Ln 1 Col 1
Type here to search
28°C Mostly cloudy
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10/22/2026

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Define charge, reference and observation points	Define charge, reference and observation points	. Correctly identifies charge distribution, point locations, distance and reference convention.	Setup is correct with minor clarification needed.	Some input or reference details are unclear.	Uses an incorrect setup or reference.	25	2.5	
2	Calculate potential and potential difference	Calculate potential and potential difference	. Calculates potential at both points and signed difference correctly with units.	. Results are mostly correct with minor arithmetic errors.	. Partial calculations require support.	Potential calculations are incorrect or absent.	25	2.5	
3	Verify potential relationship	Verify potential relationship	Checks results against analytical equations and, where relevant, the relation between E and potential.	Validation is mostly correct with minor omissions.	Validation is incomplete.	. No valid theoretical check is provided.	30	3	
4	Interpret and report findings	Interpret and report findings	. Explains the sign and physical meaning of potential difference with clear tables/conclusion	. Explanation is correct with minor omissions.	Explanation is limited or unclear.	Cannot explain or document the result.	20	2	
Total							100	10	